

No battery. PHONE: USB2 full-speed device + limited 5V input. AUX: fixed5V external input.
Global net labels connect sheets. Hardware gating defaults off.
CAD in development; no manufacturing release until ERC/DRC/BOM checks pass.

mcu_1	mcu_2	controls_1
File: mcu1.kicad_sch	File: mcu2.kicad_sch	File: controls1.kicad_sch
core_power_1	phone_input_1	aux_input_1
File: core_power1.kicad_sch	File: phone_input1.kicad_sch	File: aux_input1.kicad_sch
storage_1	storage_2	power_control1
File: storage1.kicad_sch	File: storage2.kicad_sch	File: power_control1.kicad_sch
rf_power_isolation_1	subghz_1	subghz_2
File: rf_power_isolation1.kicad_sch	File: subghz1.kicad_sch	File: subghz2.kicad_sch
nfc_1	nfc_2	lf_rfid_1
File: nfc1.kicad_sch	File: nfc2.kicad_sch	File: lf_rfid1.kicad_sch
lf_rfid_2	gpio_tools_1	gpio_tools_2
File: lf_rfid2.kicad_sch	File: gpio_tools1.kicad_sch	File: gpio_tools2.kicad_sch
ir_button_1	probe_control1	feedback_1
File: ir_button1.kicad_sch	File: probe_control1.kicad_sch	File: feedback1.kicad_sch

Sheet: /
File: FD-MAIN-01.kicad_sch
Title: FieldDock FD-MAIN-01 / Mainboard
Size: A2 | Date: 2026-09-09 | Rev: C-WIP
KiCad E.D.A. 10.0.6 | Id: 1/22



Engineering design. Production release requires full ERC, DRC, routing and BOM review.

Sheet: /mcu_1/
File: mcu_1.kicad_sch

Title: **FieldDock / mcu_1**

Size: A2 Date: 2026-09-09
KiCad E.D.A. 10.0.6

Rev: 1 / electrical review
Id: 2/22



Engineering design. Production release requires full ERC, DRC, routing and BOM review.

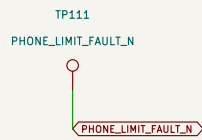
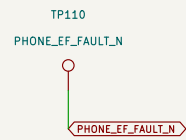
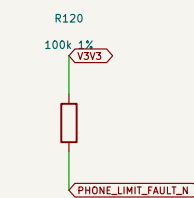
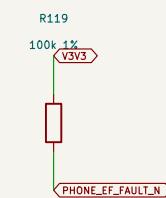
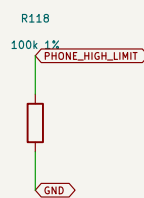
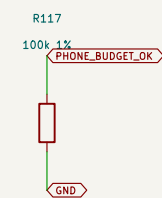
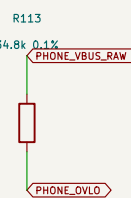
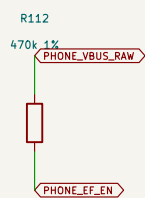
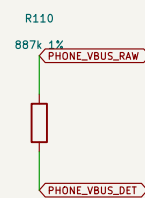
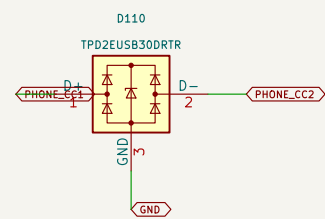
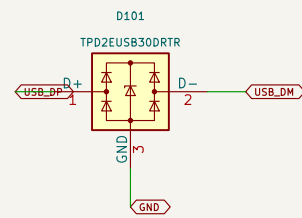
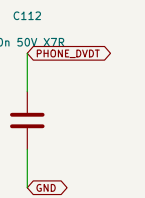
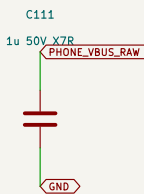
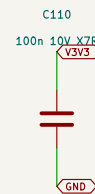
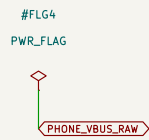
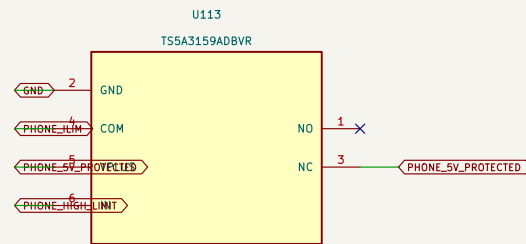
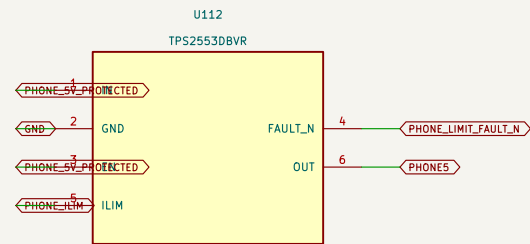
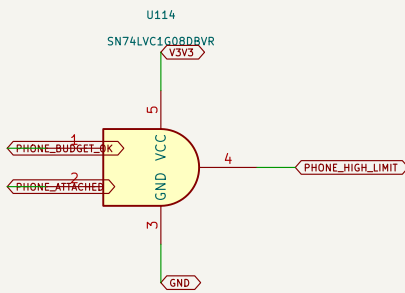
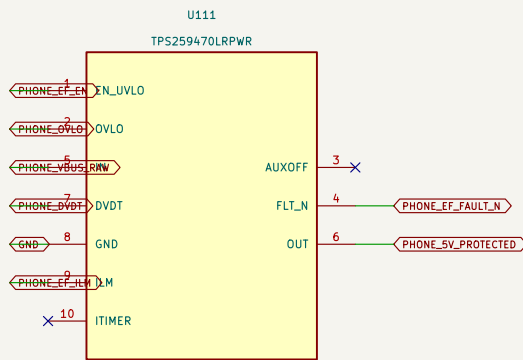
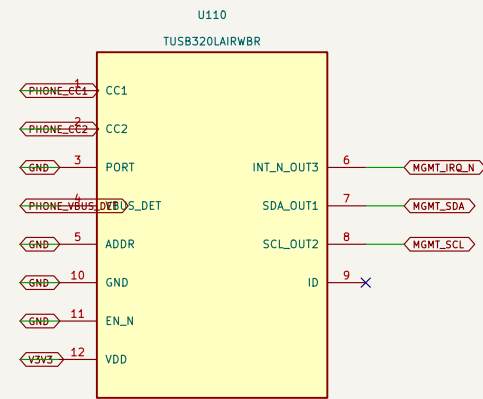
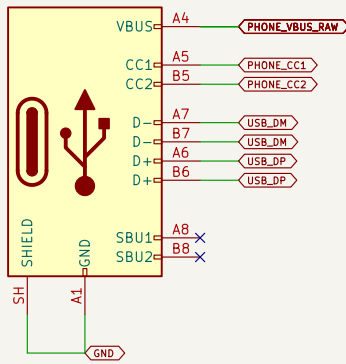
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Title: FieldDock / mcu_2		
Size: A2	Date: 2026-09-09	Rev: C-WIP
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J101
GCT USB4105-GF-A / PHONE USB2 FS + 5V



Engineering design. Production release requires full ERC, DRC, routing and BOM review.

Sheet: /phone_input_1/
File: phone_input_1.kicad_sch

Title: FieldDock / phone_input_1

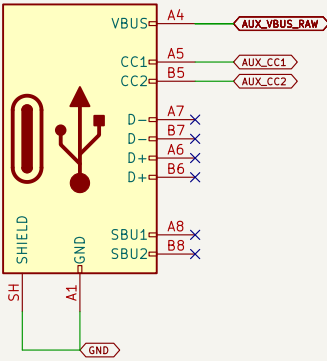
Size: A2 Date: 2026-09-09

KiCad E.D.A. 10.0.6

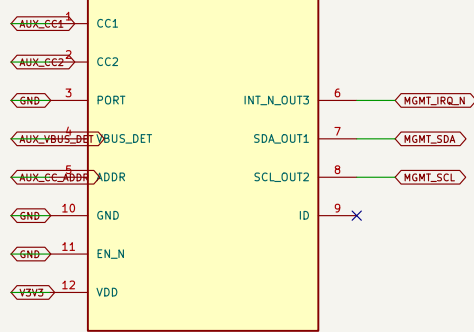
Rev: 1 / electrical review

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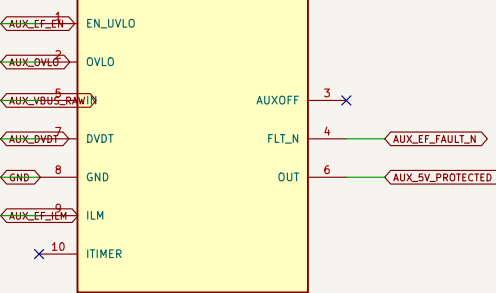
J102
GCT USB4105-GF-A / AUX 5V Input only



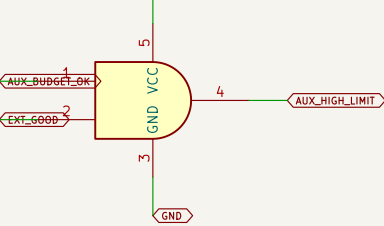
U150
TU58320LAIRWBR



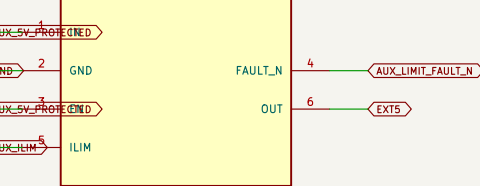
U151
TPS259470LRPWR



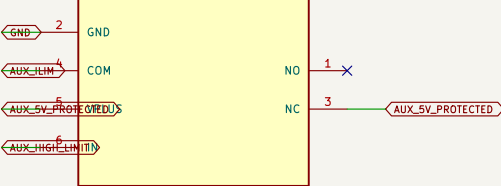
U154
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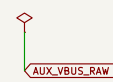
U152
TPS2553DBVR



U153
T55A3159ADBVR



#FLG5
PWR_FLAG



C150
100n 10V X7R



C151
1u 50V X7R



C152
10n 50V X7R



C153
1u 10V X7R



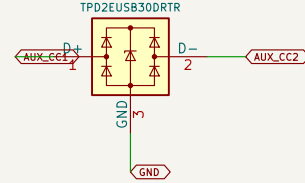
C154
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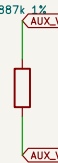
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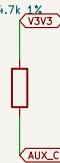
D150
TPD2EUSB300RTR



R150
887k 1%



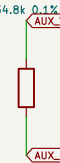
R151
4.7k 1%



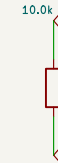
R152
470k 1%



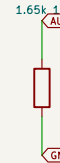
R153
3k 8k 0.1%



R154
10.0k 0.1%



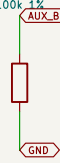
R155
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R156
3k 0k 1%



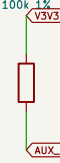
R157
100k 1%



R158
100k 1%



R159
100k 1%



R160
100k 1%



TP150
AUX_EF_FAULT_N



TP151
AUX_LIMIT_FAULT_N



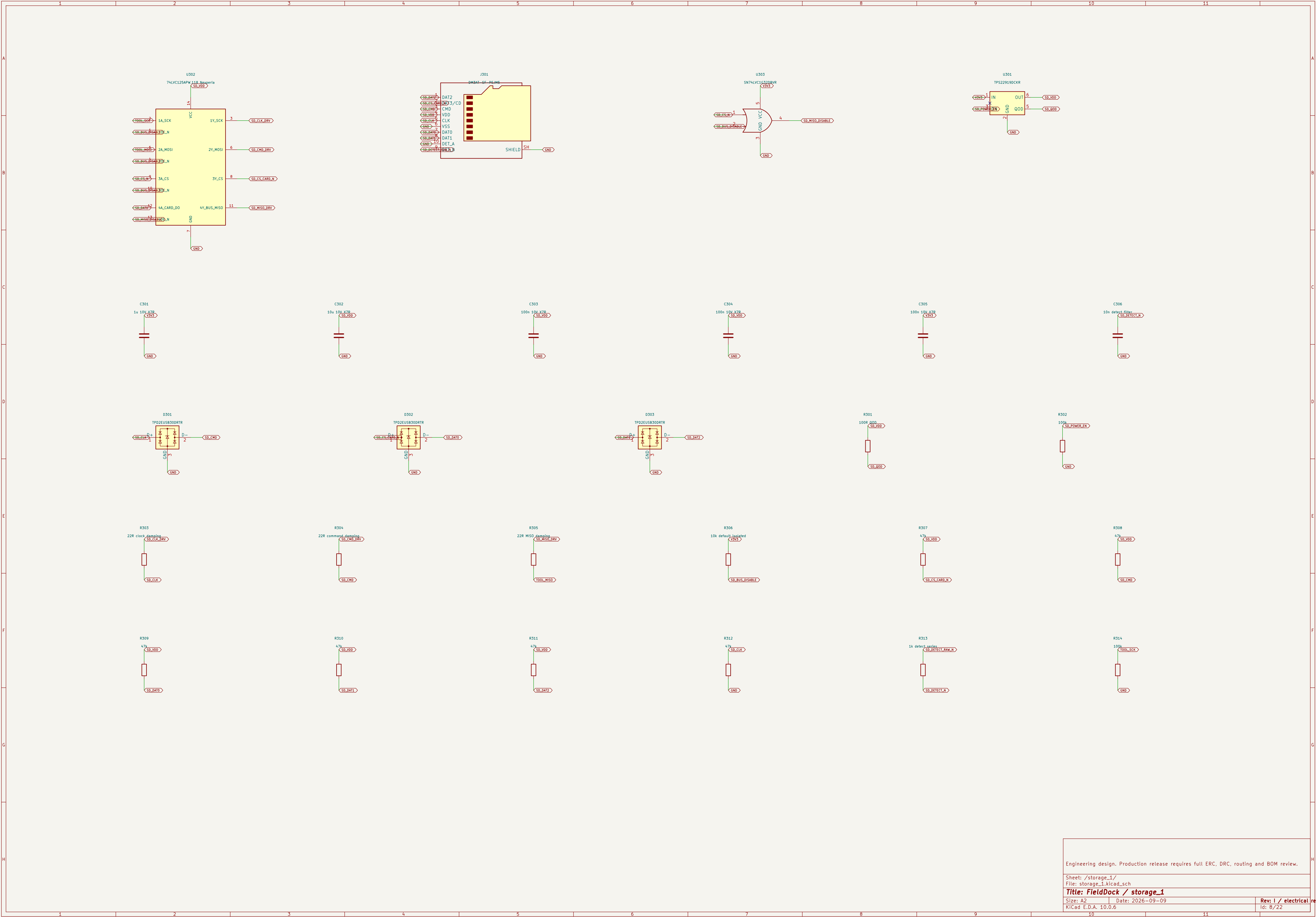
Engineering design. Production release requires full ERC, DRC, routing and BOM review.

Sheet: /aux_input_1/
File: aux_input_1.kicad_sch

Title: FieldDock / aux_input_1

Size: A2 Date: 2026-09-09
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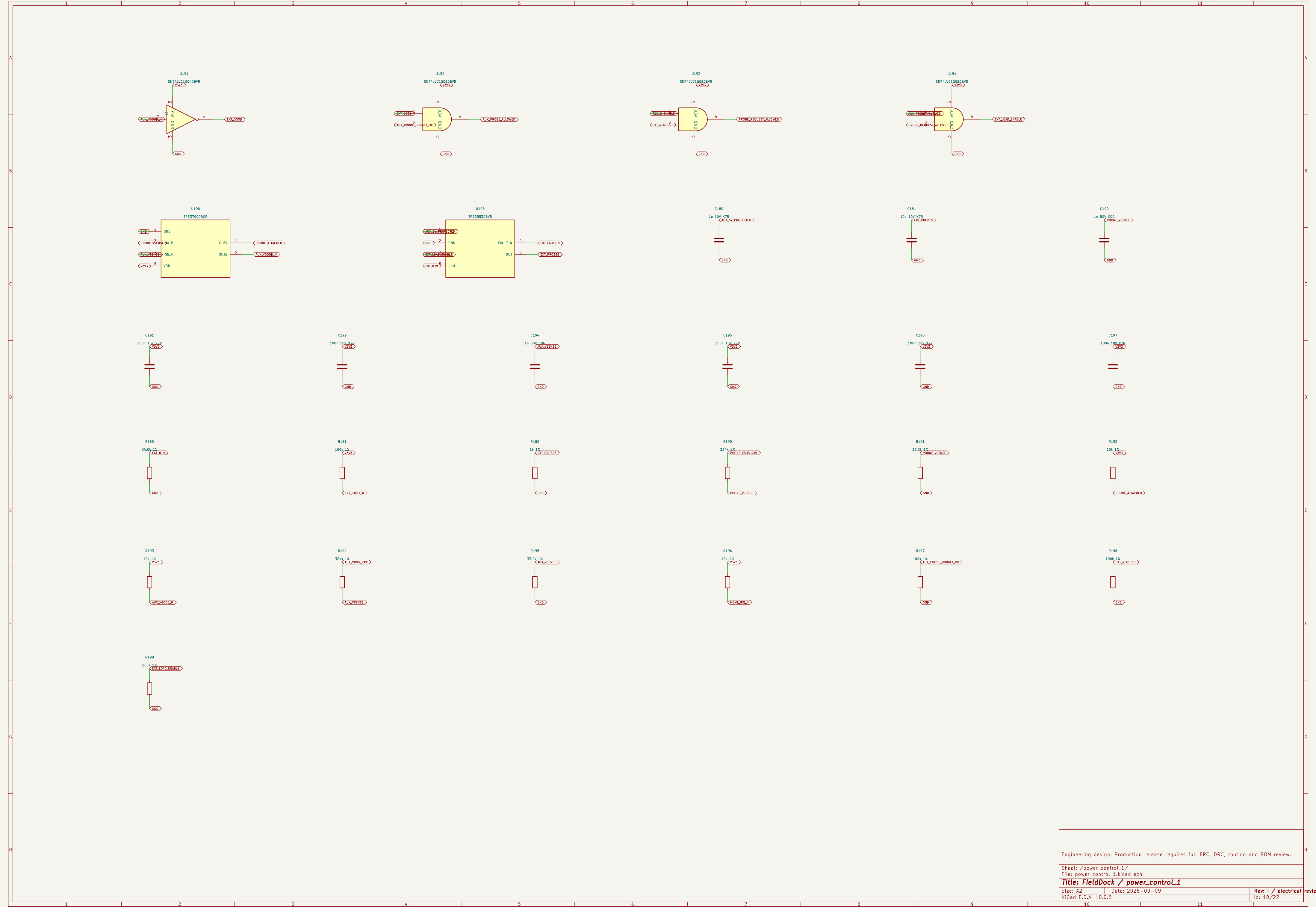


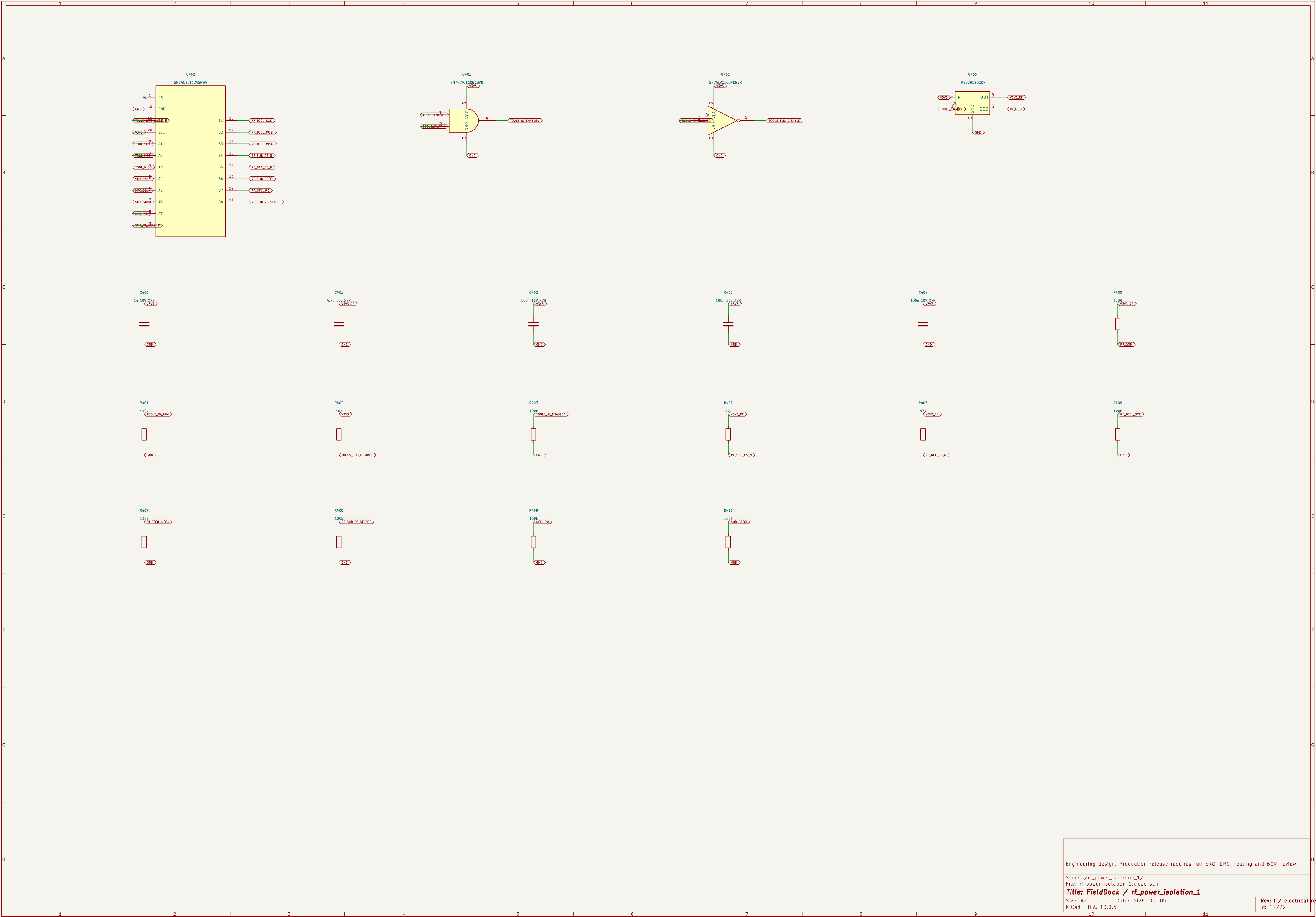


Engineering design. Production release requires full ERC, DRC, routing and BOM review.

Sheet: /storage_2/
File: storage_2.kicad_sch

Title: FieldDock / storage_2			Rev: C-WIP
Size: A2	Date: 2026-09-09		
KiCad E.D.A. 10.0.6			Id: 9/22





Engineering design. Production release requires full ERC, DRC, routing and BOM review.

Sheet: /rf_power_isolation_1/
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Title: FieldDock / rf_power_isolation_1

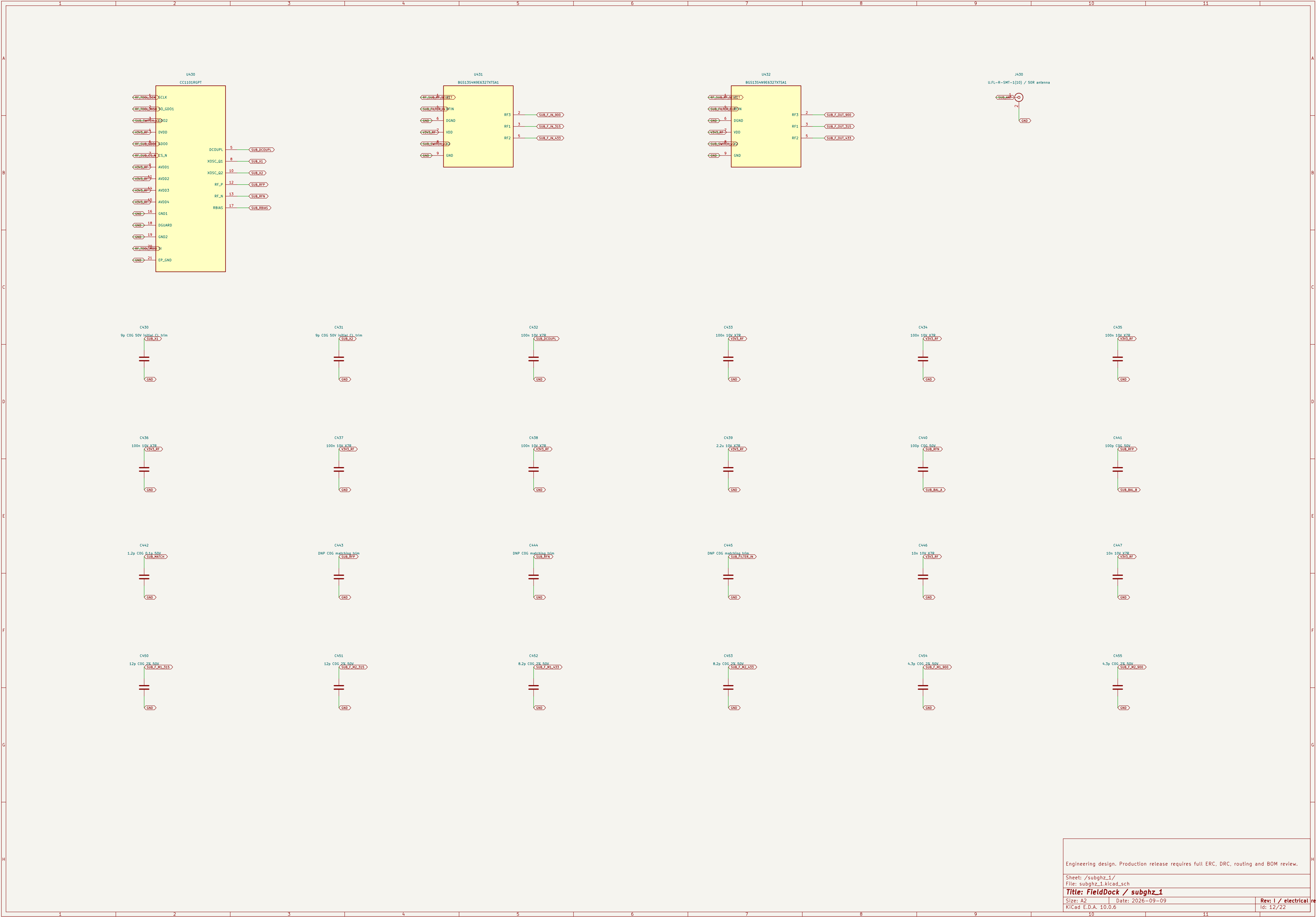
Size: A2

Date: 2026-09-09

Rev: 1 / electrical review

KiCad E.D.A. 10.0.6

Id: 11/22



Engineering design. Production release requires full ERC, DRC, routing and BOM review.

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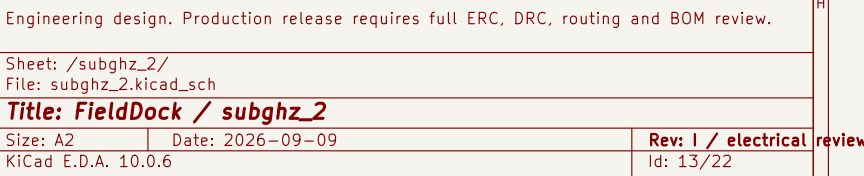
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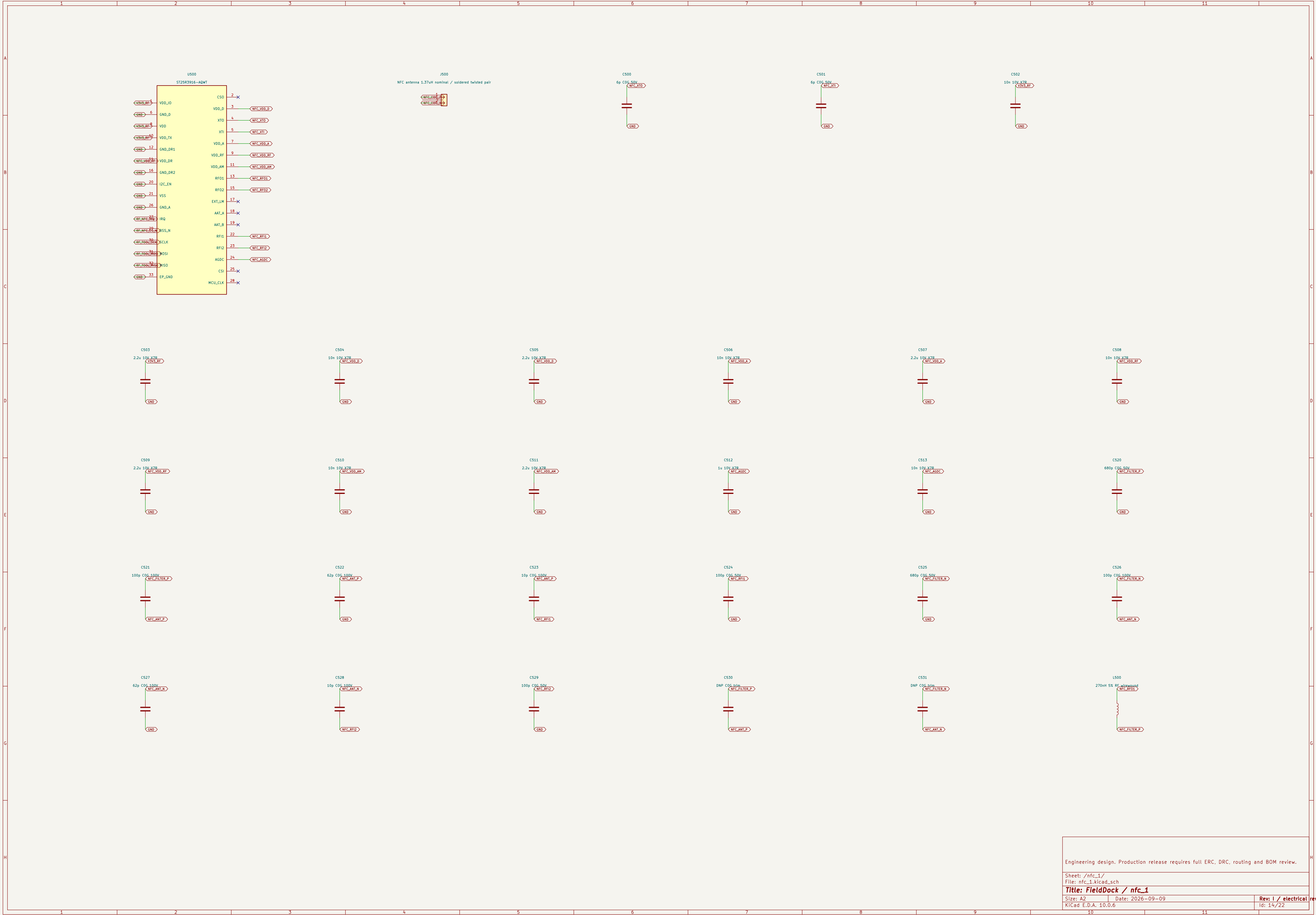
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KiCad E.D.A. 10.0.6

Rev: 1 / electrical review

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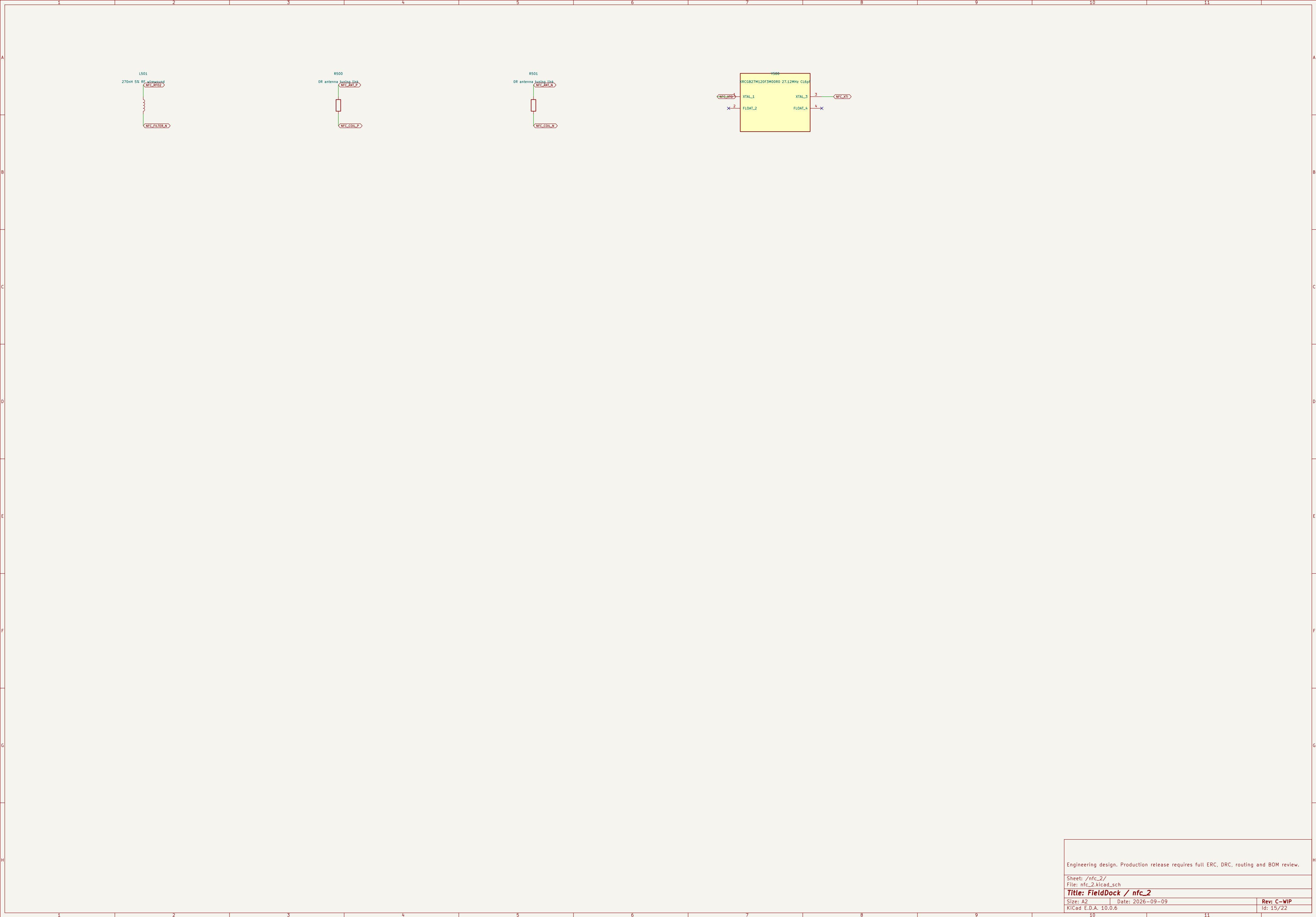


Engineering design. Production release requires full ERC, DRC, routing and BOM review.

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Title: FieldDock / nfc_1

Size: A2 Date: 2026-09-09 Rev: 1 / electrical review
KiCad E.D.A. 10.0.6 Id: 14/22



Engineering design. Production release requires full ERC, DRC, routing and BOM review.

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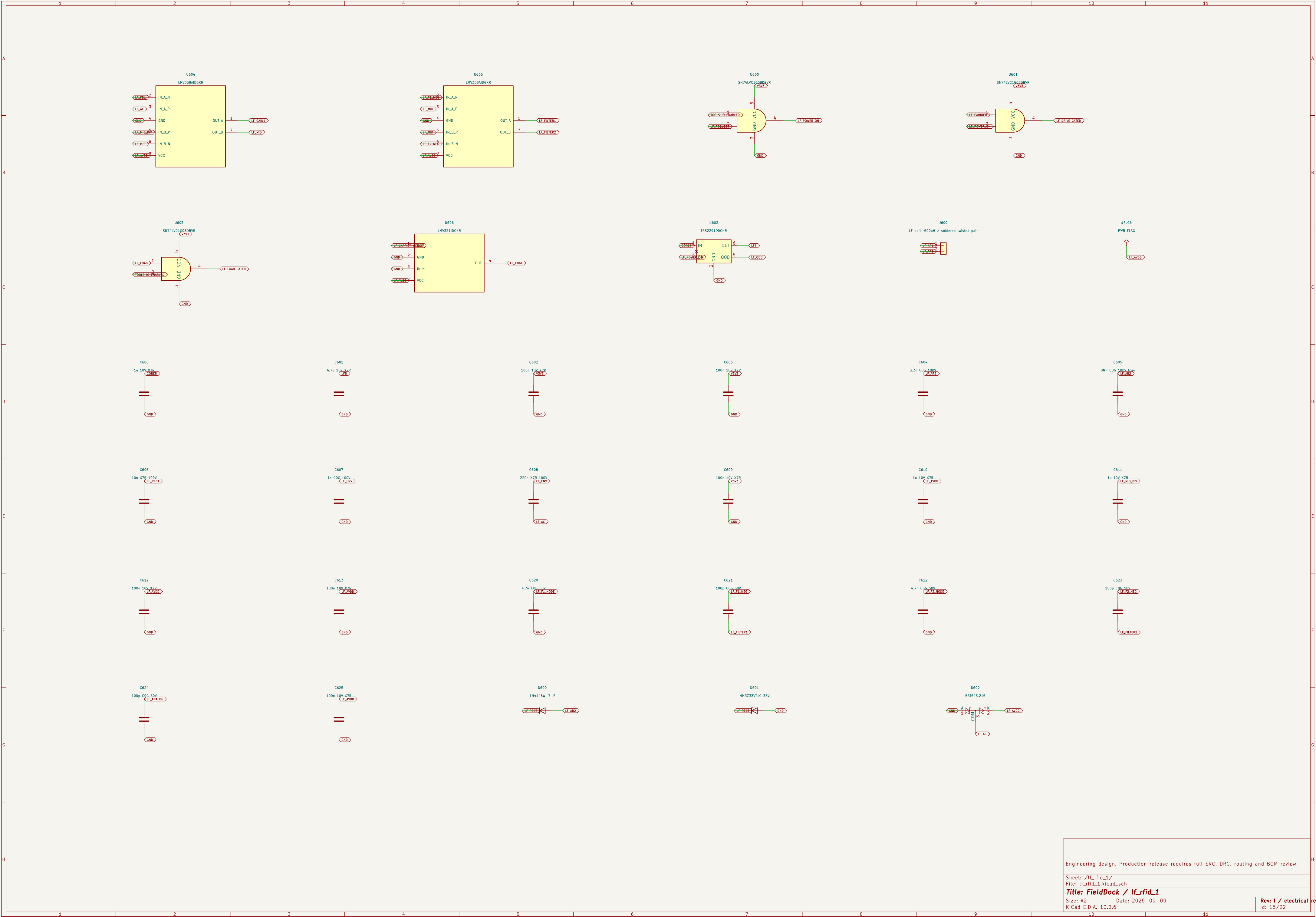
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Size: A2 Date: 2026-09-09

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Rev: C-WIP

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Engineering design. Production release requires full ERC, DRC, routing and BOM review.

Sheet: /If_rfid_1/
File: If_rfid_1.kicad_sch

Title: FieldDock / If_rfid_1

Size: A2 Date: 2026-09-09

KiCad E.D.A. 10.0.6

Rev: 1 / electrical review

Id: 16/22

